

Peyman Jabbarzade

📍 College Park, Maryland, US | ✉ peyman.jabbarzade@gmail.com | ☎ (240) 413-7609 | 🌐 pjabbarzade.github.io | in peyman-j-1566a991

Education

Ph.D. in Computer Science , University of Maryland – College Park, US	Jan 2022 – present
• Field: Theoretical Computer Science • Designing algorithms for fundamental graph problems and optimizing submodular functions in dynamic settings.	
M.Sc. in Computer Engineering , Sharif University of Technology – Tehran, Iran	Sept 2019 – Dec 2021
• Field: Algorithms and Computation • Studied geometric graph problems in the massively parallel computation (MPC) model.	
B.Sc. in Computer Engineering , Sharif University of Technology – Tehran, Iran	Sept 2015 – Aug 2019

Awards

Best Paper Award , Symposium on Foundations of Computer Science (FOCS)	Dec 2025
Dean's Fellowship , University of Maryland	Jan 2022 – Dec 2023
Asia West Champion , International Collegiate Programming Contest (ICPC) World Final	Apr 2018
14th Place , ACM International Collegiate Programming Contest (ACM-ICPC) World Final	May 2016
Bronze Medal 🏅, International Olympiad in Informatics (IOI)	July 2015
Gold Medal , Asia-Pacific Informatics Olympiad (APIO)	May 2015
Gold Medal , Iranian National Olympiad in Informatics	Sept 2014

Experience

Software Engineer Intern , Axiom – New York, US	June 2025 – Aug 2025
• Optimized circuits of the virtual machine, the core input to zero-knowledge proofs, crucial for efficient zkVM performance.	
Software Engineer , Balad (Map and Navigation App) – Tehran, Iran	Sept 2018 – Nov 2021
• Enhanced service stability and scalability and optimized inter-service communication as part of the infrastructure team. • Developed and took ownership of Balad's core service for managing and providing location-specific data. • Led a diverse team of 10 engineers to collect and clean location-specific data, and to refine the service design using microservices. • Primarily coded in Python, with additional experience in C++, Java, Go, and various other tools.	
Research Intern , Max Planck Institute for Informatics – Saarbrücken, Germany	July 2019 – Aug 2019
• Designed and benchmarked an energy-preserving online scheduling algorithm.	
Software Engineer Intern , Balad (Map and Navigation App) – Tehran, Iran	June 2018 – Aug 2018
• Designed and implemented a novel algorithm for optimal routing of camera-equipped cars, ensuring comprehensive street coverage in Tehran within the minimum time.	

Publications

(All papers use **alphabetical** author ordering.)

Breaking a Long-Standing Barrier: $2-\epsilon$ Approximation for Steiner Forest 🏅	Dec 2025
Ali Ahmadi, Iman Gholami, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Mohammad Mahdavi 66th Annual Symposium on Foundations of Computer Science (FOCS 2025, best paper award).	
Dynamic Diameter in High-Dimensions against Adaptive Adversary and Beyond	Dec 2025
Kiarash Banihashem, Jeff Giliberti, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh 39th Conference on Neural Information Processing Systems (NeurIPS 2025).	
Dynamic Algorithms for Non-monotone p-Matchoids	Dec 2025
Kiarash Banihashem, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh 39th Conference on Neural Information Processing Systems (NeurIPS 2025).	
Dynamic Algorithms for Submodular Matching 🏅	July 2025
Kiarash Banihashem, Leyla Biabani, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh	

52nd EATCS International Colloquium on Automata, Languages, and Programming (**ICALP 2025**).

Prize-Collecting Forest with Submodular Penalties: Improved Approximation [↗](#)

Jan 2025

Ali Ahmadi, Iman Gholami, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Mohammad Mahdavi

26th Conference on Integer Programming and Combinatorial Optimization (**IPCO 2025**).

A Dynamic Algorithm for Weighted Submodular Cover Problem [↗](#)

July 2024

Kiarash Banihashem, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh

41st International Conference on Machine Learning (**ICML 2024, accepted for oral presentation**)

Prize-Collecting Steiner Tree: A 1.79 Approximation [↗](#)

June 2024

Ali Ahmadi, Iman Gholami, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Mohammad Mahdavi

56th ACM Symposium on Theory of Computing (**STOC 2024**).

2-Approximation for Prize-Collecting Steiner Forest [↗](#)

Jan 2024

Ali Ahmadi, Iman Gholami, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Mohammad Mahdavi

Journal of the ACM (**JACM 2025**). ACM-SIAM Symposium on Discrete Algorithms (**SODA 2024, selected for special issue**).

Dynamic Algorithms for Matroid Submodular Maximization [↗](#)

Jan 2024

Kiarash Banihashem, Leyla Biabani, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh

ACM-SIAM Symposium on Discrete Algorithms (**SODA 2024**).

Dynamic Non-monotone Submodular Maximization [↗](#)

Dec 2023

Kiarash Banihashem, Leyla Biabani, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh

37th Conference on Neural Information Processing Systems (**NeurIPS 2023**).

Dynamic Constrained Submodular Optimization with Polylogarithmic Update Time [↗](#)

July 2023

Kiarash Banihashem, Leyla Biabani, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh

40th International Conference on Machine Learning (**ICML 2023**)

A Novel Prediction Setup for Online Speed-Scaling [↗](#)

June 2022

Antonios Antoniadis, Peyman Jabbarzade, Golnoosh Shahkarami

18th Scandinavian Symposium and Workshops on Algorithm Theory (**SWAT 2022**)

Volunteer Experience

Paper Reviewing

2020 – present

- Junior Program Committee member, SPAA 2025.
- Program Committee member, AAAI 2026.
- Reviewer for Algorithmica journal and conferences such as DISC 2022, ALGOSENSORS 2022, NeurIPS 2024, IPCO 2024, APPROX 2024, ESA 2024, ICML 2025, ICLR 2025, AISTATS 2025, ICALP 2025, SPAA 2025, NeurIPS 2025, and AAAI 2026.

Coaching

2020 – 2023

- Coached the University of Maryland's team for ICPC 2023, leading them to a silver medal at the North America Championship and advancing to the world final. [\[1 ↗, 2 ↗\]](#)
- Led [Iran's national team for IOI 2020 ↗](#), achieving 3 gold and 1 silver medal—the best result for Iran in IOI.
- Coached Sharif University of Technology's team in the ICPC 2020 World Finals.

Scientific Committee

2017 – 2024

- Member of the Host Scientific Committee (HSC) for IOI 2017.
- Judge for the ICPC North America Championship 2024.
- Member of the Iranian National Olympiads in Informatics Scientific Committee over six years.
- Contributed to the preparation of the ICPC Tehran Regional Contest over six years.

Teaching

2015 – 2024

- Teaching Assistant at the University of Maryland for courses in Algorithms, Advanced Data Structures, and Computational Genomics.
- Teaching Assistant at Sharif University of Technology for courses in Design Algorithms and Discrete Structures.
- Prepared high school students for the Iranian National Olympiad in Informatics.

Task Author

2015 – 2024

- Authored the problem "[Rectangles ↗](#)" for [IOI 2019 ↗](#).
- Managed numerous programming competitions and authored problems for selecting Iran's national team for IOI.

Technical Skills _____

Programming Languages: Python, C/C++, SQL

Technologies: Docker, Kubernetes, Kafka, gRPC, REST, Sentry, Redis, Prometheus, Grafana, ElasticSearch, Logstash, Kibana